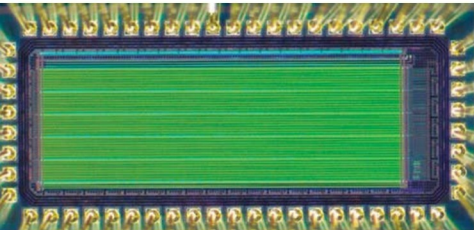


CAMs: MEMORIES FASTER THAN RAMs

Content-addressable memories (CAMs) are useful for applications like networking where search time must be very short



High-speed content-addressable memory using synchronous hamming distance search circuits

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Content-addressable memories (CAMs), in their most trivial form, work in a way opposite to conventional memories. While a random-access memory (RAM) returns data upon providing the address, a CAM provides address (which in this context is commonly called the index) when data is supplied. So effectively a CAM is a search tool that looks for the data in question within its database.

CAMs commonly work as hardware search engines in routers and switches to accelerate forwarding of packets on the destination route. We will talk about the various other applications that consider a CAM as an important processing engine as we proceed in this article.

As indicated in its definition, CAM holds the addresses (IP or MAC) to be looked-up for and returns the route-associated information to the host processor to perform the successive tasks of forwarding the incoming packet. So we can also define a CAM as a network coprocessor to the central processor.

Note that TCP/IP and/or OSI models are taken here as reference to explain the concepts; similar analogies may be drawn with other protocols like IPX/SPX, AppleTalk and UDP.

Functionality of CAMs

In order to get a true understanding of the functionality of CAMs, you need to dig a little deeper into the domain in which these work. If you look at either the OSI reference model or the TCP/IP stack, you'll find that the

lower three layers are the ones that are concerned with the packet delivery to its peer, while all the other layers are more concerned with the content, its presentation and formatting.

The physical layer adopts analogue means of data transfer and is not concerned with acknowledgement of the transmitted data. This leaves the next two layers (data link layer and IP/network layer) to make sure that the transmitted packet was indeed received in its original form—that is, the order of information, encryption used for data protection, etc.

Another important task that needs to be performed by these upper layers is to ensure delivery of the packet to its correct recipient. This is done in two ways:

1. *Using MAC (media access control) address.* This addressing scheme is adopted at Layer 2 of the OSI layer. It is used in such network devices as switches/bridges.

2. *Using IP addresses.* This addressing scheme is used to transmit any IP packet and communicate with its peer on the network, which is the data link layer on the destination (Layer 3 of the OSI stack). It is used in such network devices as routers and Layer 3 switches.

Building a CAM

When a forwarding device receives a packet on its ingress port, based on the layer at which the device is working (Layer 2/Layer 3), the device looks up for the corresponding address in